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RESEARCH ARTICLE

Enhancing Urban Heatwave Response Planning via a Graph-Based Digital Twin Approach: Spatial Dependency Risk Analysis in Vienna City

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ABSTRACT Urban heatwaves pose an escalating threat to cities, requiring advanced planning strategies. This paper presents a graph-based digital twin approach to model spatial dependencies in heatwave risk for urban areas. Using Vienna's city center as a case study, we integrate (i) high-resolution spatial data from Google on heat hazard, (ii) spatial population distribution from Vienna Open Data and (iii) META's Vienna population datasets alongside (iv) critical road infrastructure from OpenStreetMaps, into a graph network model that captures cumulative risk paths that help determine the most critical city blocks needing intervention and the optimal allocation of resources for routing first responders to reach critical locations of sensitive populations. Our Spatial Dependency Risk Analysis identifies how the distribution of vulnerable population needing first aid in one location together with heat data may amplify risks in connected areas and provide a planning tool to first responders to draft optimal routes to reach all persons in need as fast as possible. The results show how city center junctions form interdependent high-risk clusters; notably, our tool managed to identify the most at-risk sub-district reaches and automatically plan ambulance routes from cooling spots to high risk hotspots. To our knowledge, this is the first automatic tool able to quantify these network routing effects during heatwave disasters, and enhance heatwave response planning by pinpointing critical areas where targeted interventions (e.g., cooling centers or outreach) could substantially reduce cascading impacts

INDEX TERMS Risk assessment, digital twin, smart city, critical infrastructure, heatwave, first responders.

I. INTRODUCTION

Prolonged and intense heat waves severely affect society, especially urban environments ([1], [2], [3]). Urban heatwaves pose a significant threat to public health, especially in densely populated areas where vulnerable groups are at increased risk. These events place unprecedented stress on critical services, especially healthcare services and fire departments. In this context, health services have been identified by several government agencies and authorities (e.g., Cybersecurity and Infrastructure Security Agency, the European Union with the NIS2 Directive[1]) as critical entities.[2] Effective preparedness and response planning are vital to mitigating the impact of such extreme events.

To address the significant risks posed by the increasing number of prolonged heat events, we build a digital

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[1]https://eur-lex.europa.eu/legal-content/en/TXT/?uri=CELEX%3A32022L2555

[2]https://www.cisa.gov/topics/critical-infrastructure-security-and-resilience/critical-infrastructure-sectors/healthcare-and-public-health-sector

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twin simulation of heatwaves over an urban network of city blocks that is able to prioritize critical locations and automate resource allocation and route planning for first responders. A Digital Twin (DT) is formally defined in ISO 23247-1:2021 as a ''fit for purpose digital representation of an observable manufacturing element with a means to enable convergence between the element and its digital representation at an appropriate rate of synchronization'' [35]. The methodology presented utilizes satellite data provided by the Google Solar API[3] for real-time heat index calculations to reveal areas within cities that are more exposed to extreme temperatures. Then, it correlates produced Heat Risk maps with population data, such as density and vulnerability of citizens (e.g., age group below 14 and above 60, as per the IPCC recommendations[4]), to identify the most at-risk urban city blocks and propose routing paths and efficient station points that can be utilized by emergency teams and first responders for faster or more frequent itineraries.

A. CONTRIBUTION

In this work, the systematic integration of satellite-derived heat exposure data, demographic vulnerability metrics, and network dependency analytics for emergency response planning are combined to introduce the concept of data-driven Spatial Dependency Risk Analysis for planning responses of first responders.

The case study of Vienna's city center demonstrates the method's effectiveness in a real-world context, the methodology itself is designed independently of specific location, ensuring its applicability to diverse urban settings or other extreme weather scenarios. Its generalizability is supported by reliance on commonly accessible data sources, standardized graph-based analyses, and reproducible AI-driven risk modeling approaches.

The digital twin's objective is to map areas of vulnerable population in Vienna's city center against solar radiation data and combine them with graph and centrality analytics regarding road junctions in the same areas of interest. This way, first responders can identify paths that will cover the majority of high-risk locations – locations that have been identified as essential and where many vulnerable populations reside – or plan for relocation of specific points/areas or parts of facilities and identify the nominal road routes.

We apply a road network analysis methodology using graph centrality [4] [26] [27] and dependency risk chains [26] and extend it to the domain of heat index analysis. We then used it to create a digital twin of the Vienna, Austria city center to identify critical urban blocks and prioritizing response strategies during heatwaves. This research is a significant contribution to the field, achieved by integrating data from the Google Solar API for real-time heat index calculations with population distribution data obtained from the Vienna Open Database [5]. Our main contributions are:

1) A smart city digital twin for first responders able to model the criticality and prioritization of urban blocks and respective road junctions as nodes, where each node is characterized by its heat risk index of the corresponding city blocks in the vicinity. By analyzing this dependency graph, we can calculate cumulative risk paths that help determine the most critical blocks needing intervention and the optimal allocation of resources for first responders. We can also detect the shortest path to reach locations of sensitive populations, considering heat exposure and population density. The aforementioned centrlity metrics effectively capture network vulnerability and, even though they are widely used within the domain of transportation network analysis and thus, alone, do not represent methodological novelty, still to the best of our knowledge, these metrics are not used before in determining node importance for urban areas in conjunction with heat indexes and traffic analysis.
2) The proposed method models risk, in this case, extreme heat events (e.g., heatwaves) and vulnerable assets (e.g., population and infrastructure), over a transportation network able to allocate resources such as ambulances and cold spots. Our model can serve as a risk identification and management tool by combining heat exposure data (provided by the Google Solar API) with demographic information, pinpointing the most critical areas in urban environments. In that view, it assists city authorities and first responders in planning for heat-wave scenarios more effectively, maximizing coverage of high-risk zones within minimum response times.
3) Another essential contribution of our proposed methodology is that it enables the merging of population protection with risk management theory, ensuring that equity is built into emergency response systems and processes by design and by default. Vulnerable populations often face higher risks not only because of physical exposure but also due to socio-economic factors that limit their access to healthcare and other critical resources during emergencies.

While demonstrated through the Vienna case study, these optimizations are applicable to other urban settings, requiring only localized adjustments in demographic and environmental data. The methodology is inherently adaptable to other cities and diverse extreme events due to its modular design, reliance on standardized APIs, and open data sources. OpenStreetMaps, Google Solar API and open government population data, including vulnerable populations, are all inherently available input and all that is necessary to setup the digital twin simulation for any city and provide analytics for planning the routing of emergency vehicles, and strategic allocation of limited emergency resources (such as ambulances, cooling stations, or temporary relief centers).

The remainder of this paper is organized as follows. The next section reviews existing literature relevant to urban

[3]https://developers.google.com/maps/documentation/solar/overview
[4]https://www.ipcc.ch/report/ar6/wg2/

heatwave impacts, transportation risk modeling, and graph-based dependency analyses. In Methodology, we detail the procedures used for data acquisition, vulnerability analysis, risk assessment calculations, and graph centrality metrics utilized within the digital twin model. Subsequently, the Results section presents the outcomes from applying this methodology to the case study of Vienna's city center, including critical nodes identification, dependency paths, and optimal response routes. Lastly, a dedicated Implications and Future Work section addresses the implications of our method for urban planning and references public policy alignment with EU regulations, and emergency preparedness and lays down future research directions.

II. RELATED WORK

A. URBAN HEATWAVE IMPACT

In urban and highly populated areas, such as cities and metropolises, the heat waves are further impacted due to the urban heat island (UHI) effect [6], which indicates a significantly larger temperature than the surrounding rural areas. Typically, the temperature difference is higher at night than during the day and is most apparent when the wind weakens. The main contributor to the UHI effect is the modification of the land areas to large blocks of concrete buildings, with secondary contributions arising from several factors, such as heat generated by energy consumption used for air conditioning purposes [6]. More than half of the Earth's population lives in urban areas, which occupy only 0.5% of the land surface [7]. The heat wave magnitude index (HWMI) proposed by [8] led to the possibility of comparing heat waves (or extreme heat events) in different regions of the world with different climates. This approach provides a foundation for urban heat risk management. In [23] several tools, such as satellite data and GIS tools, have been used to create a comprehensive urban heat vulnerability map – in this case for the city of Vienna, Austria – which integrated factors such as exposure, sensitivity, and adaptive capacity.

B. ROAD TRANSPORTATION RISK MODELS

Transportation networks play a pivotal role during heatwave emergencies, particularly in urban areas or city centers where traffic congestion can severely affect the rapid response of emergency teams. Several road network simulation models have been introduced to mathematically model the transportation systems, which are too complicated for simple analytical or numerical analysis. These models are based on the concept of system state, a set of variables that describe the system state over time [9]. System state can be either discrete or continuous. The more recent methods utilize either the discrete event simulation (a discrete series of events over time) or the continuous-time simulation (a set of continuously changing variables). We can identify two distinct prediction models: the short-term, which should be precise in providing the road network characteristics in the case of an adverse event, and the long-term, which provides a broader road network picture over time but with limited accuracy in abnormal or atypical events. Enhancing critical road segments, especially in post-disaster recovery actions, can be achieved by using graph neural networks [28]. Furthermore, network improvements and effective management strategies can be achieved using several centrality metrics (such as betweenness centrality) [29] to identify critical nodes in transportation networks. In addition, web-based decision support systems, as suggested in [30], can assist in strategically placing emergency facilities to support accessibility options during disasters or adverse situations. In heatwave scenarios, optimizing traffic flow becomes critical, particularly for routes connecting areas with high-risk populations.

Our work demonstrates how spatial dependency risk analysis can be used to identify optimized response paths that account for dependencies among urban zones that host vulnerable populations during extreme events. To our knowledge, there is currently no solution capable of calculating cumulative risk paths that help determine the most critical areas needing intervention and the optimal allocation of resources for first responders and detect the shortest path to reach critical locations of sensitive populations, considering heat exposure and population density.

The proposed digital twin methodology significantly extends existing GIS-based emergency response models by integrating real-time solar radiation data, population demographics, and spatial dependency analyses based on network centrality metrics. Existing GIS methodologies primarily are focused on vulnerability mapping.

We have created a comparative table that depicts various characteristics reported in related literature alongside the capabilities demonstrated in our study. For example, graph-based digital twins leverage integrated, real-time data streams (IoT sensors, dynamic climate inputs, etc.), whereas classical GIS models typically rely on static map layers mdpi.com. Our digital twin approach is inherently time-aware (able to simulate historic, current, and future states) esri.com and can model interdependencies and scenario dynamics iccgis2024.cartography-gis.com beyond the static ''snapshot'' analyses of conventional methods.

Table 1 highlights these differences in data sources, temporal resolution, spatial granularity, and analytic functionality. Our approach enables predictive modeling of cascading impacts across interconnected urban zones, significantly enhancing the precision and practical applicability of emergency planning. Additional references highlighting this comparison have been included in the literature review for greater clarity. Unlike traditional GIS-based approaches [8], [23], which typically generate static vulnerability assessments, our method dynamically evaluates cumulative dependency risks, providing insight for prioritization of critical areas and optimized routing.

III. BACKGROUND

The proposed methodology for building the digital twin leverages multiple previously proposed modeling techniques [4],

TABLE 1. Comparison of the proposed digital twin-based spatial risk analysis vs. classical GIS-based emergency models.

Aspect	Graph-Based Digital Twin (Proposed)	Classical GIS-Based Emergency Model
Data Sources	Integrates diverse data (GIS layers, real-time IoT sensors, BIM, and dynamic climate inputs), updating continuouslymdpi.com.	Relies on static GIS data layers (e.g., historical hazard maps, census demographics) with infrequent updates.
Temporal Dynamics	Time-aware and dynamic – can simulate evolving scenarios and update in near real-timeesri.com. Supports historic, current, and forecast state analysis.	Largely static or scenario-specific snapshots. Temporal analysis is limited (often one-off simulations or past event assessments).
Spatial Resolution	Fine-grained representation (e.g., graph nodes at block or building level). Captures local inter-node relationships explicitly.	Coarser aggregation (e.g., city districts or grid cells). Interdependencies between areas are usually not explicitly modeled.
Analytic Capabilities	Simulates interdependencies and cascading effects in the urban system; allows "what-if" scenario testing and network metrics (connectivity, centrality) for risk propagationiccgis2024.cartography-gis.com.	Primarily overlay and proximity analysis; identifies static hotspots and performs routing or allocation tasks, but cannot easily simulate chain reactions or system-wide feedback in real-time.

[26], [27], and uses them to build an assessment model for determining vulnerability and routing in transportation infra-structures by analyzing the inherent vulnerability of city blocks and the dependency chains between them using the road networks. Junction interdependencies are modeled exclusively at the node level, focusing on risks associated with city blocks or junctions rather than traffic flow or road interconnections. Junction interdependencies are considered as roads or highways as edges that connect city junctions.

Using a graph-based approach, high-risk dependency paths – routes where congestion at a single junction can have a cascading effect on the whole network – can be identified. The research emphasizes that understanding the n_{th}-order dependencies between junctions is crucial for forecasting cascading effects in urban infrastructure systems. The presented methodology leverages on a CI dependency analysis method [26], which uses statistical methods and graph theory to model dependencies and propose risk-based mitigation actions. This method was chosen since graphs are a very useful isomorphism to the road network. Still, many different approaches exist in model infrastructure, for example and within the context of critical infrastructure protection at a level like a large metropolis such as Vienna city, and the legal attributions of cyber operations, several interdisciplinary methodologies have been introduced. For example, Pipyros et al. [38], proposed a novel model leveraging Multiple Attribute Decision Making (MADM) techniques in conjunction to the Tallinn Manual's criteria to evaluate the scale and effects of cyber operations. In this regard, our proposed digital twin framework complements and extends such approaches by enabling dynamic, real-time simulations of cyber-induced disruptions within complex urban infrastructures.

A. DEPENDENCY ANALYSIS METHODOLOGY

In discrete mathematics, a graph is a collection of objects that are represented by abstractions called vertices/nodes, and each of the related pairs is called an edge in the sense that some pairs of the objects are related [10]. The number |V(G)| of its vertices is called the order of the graph. Graphs can be either *undirected* or *directed*. In an *undirected graph*, edges have no direction, meaning the relationship between two vertices is bidirectional. Dependency analysis assesses the risk of *n*th-order dependencies by applying results of organization-level risk assessments. We define "dependency" as a "*one-directional reliance of an asset, system, or collection thereof on an input, interaction or other requirement from other sources to function properly*" [11], [12].

In our case, *dependency risk* is quantified at the node level, representing the risk associated with urban junctions or city blocks. The overall risk of a path is then derived as the cumulative aggregation of the risks of the nodes along that path. Each node N_i is assigned to an individual risk R_{N_i}, which is calculated based on its likelihood and impact factors as per the following formula (1):

$$R_{N_i} = L_{N_i} \cdot I_{N_i} \tag{1}$$

where L_{N_i} represents the likelihood of the node experiencing a risk, and I_{N_i} quantifies the impact at that node.

For a path $P = \{N_0, N_1, \ldots, N_k\}$ with N_0 as the source node, the *cumulative dependency risk* for path P is expressed (2) as the summary of all the risks at each node along the route:

$$CDR_P = \sum_{i=0}^{k} R_{N_i} \tag{2}$$

In the above formula, CDR_P represents the total risk of the path, as calculated by the aggregation of all the risks of the nodes that make up the path. This methodology provides the ability to tailor results based on path length, defined as the number of nodes (or city blocks) included in the path. In that way, we can identify the most critical routes for the specific path length selected. This means that, for a given path length k, the methodology provides a set of city blocks (represented as nodes) that collectively have the highest cumulative risk. In that view, we provide first responders and digital twin users with the ability to adapt the methodology to different emergency response strategies and to adjust the desired path length to create optimized paths for several operational objectives, such as:

Shorter paths: where rapid response and quick access to a subset of critical nodes are essential.

Longer paths: where comprehensive coverage of high-risk areas over an extended route is required.

B. CENTRALITY METRICS

Within graph theory, a plethora of measures have been proposed, especially within the context of transportation [27], to analyze complex networks. We can identify *centrality* as a quantitative measure to assess a node's influence, significance, or importance within a network (graph). The term ''importance'' can be defined and measured from different perspectives, depending on the context and the specific metric used. These metrics are based on specific criteria, such as the number of direct connections (degree centrality), closeness to all other nodes (closeness centrality), etc.

PageRank, Betweenness and Eigenvector metrics were selected based on their suitability to reflect diverse dimensions of urban networks [4], [27]. Betweenness centrality identifies critical nodes functioning as vital connectors essential for maintaining emergency route accessibility [4]. Eigenvector and PageRank metrics were selected to quantify nodes' cascading influences and their roles network wide [27]. Computationally, these centrality algorithms scale polynomially (O(V^2) to O(V^3), where V is the number of nodes), confirming their suitability for application to city-scale transportation networks.

The aforementioned centrality metrics effectively capture network vulnerability and, even though they are widely used within the domain of transportation network analysis and thus, alone, do not represent methodological novelty, still to the best of our knowledge, these metrics are not used before in determining node importance for urban areas in conjunction with heat indexes and traffic analysis.

Below, we provide a short analysis of each centrality metric:

1) Degree centrality, conceptually the simplest centrality measure, measures a node's importance based on the pro-portion of edges a node has to the maximum possible number of edges in the network (graph) [13]. Nodes with more connections to other nodes are considered more central and, therefore, will be measured with a high degree of centrality. In a graph where N is the total number of nodes, the number of edges connected to a node v, represent the degree centrality $C_{D(v)}$, as per the following formula:

$$C_{D(v)} = \frac{degree\,(v)}{N-1} \tag{3}$$

2) Betweenness centrality is first introduced as a general measure applicable to a wide range of network theory problems, such as social networks and transportation. For every pair of vertices in a connected graph, at least one shortest path exists between the vertices such that either the number of edges that the path passes through (for un-weighted graphs) or the sum of the weights of the edges (for weighted graphs) is minimized. The betweenness centrality for each vertex is the number of these shortest paths that pass through the vertex [14]. In other words, betweenness centrality quantifies the extent to which a node acts as a bridge to other nodes, thus controlling the flow of information. As per Freeman [15], who gave the first formal definition, the betweenness centrality $C_B\,(i)$ of a node i is given by the expression:

$$C_B\,(i) = \sum\nolimits_{j<k} \frac{\sigma_{jk}\,(i)}{\sigma_{jk}} \tag{4}$$

where, σ_{jk} is the total number of shortest paths from node (j) to node (k), and $\sigma_{jk}(i)$ is the number of those paths that pass through (i).

3) Closeness centrality, how close a node is to all other nodes in a network, is calculated as the reciprocal of the sum of the length of the shortest paths between the node and all other nodes in the graph [16]. In that view, nodes with high closeness centrality can quickly interact with other nodes, effectively spreading information throughout the network. The closeness centrality $C_C\,(i)$ of a node i is defined as the reciprocal of the sum of the shortest path distances from i to all other nodes [17]:

$$C_C\,(i) = \frac{1}{\sum_{j\neq i} d\,(i,j)} \tag{5}$$

Typically, the normalized version of the above formula is used, representing the average length of the shortest paths instead of their sum.

$$Normalized\colon C'_C\,(i) = \frac{N-1}{\sum_{j\neq i} d\,(i,j)} \tag{6}$$

4) Eigenvector centrality [18] measures the influence of a node in a connected network based on the importance of its neighbors or identifies a node that is connected to several other influential nodes, which can amplify their impact. Eigenvector centrality is commonly applied to identify influential individuals in social networks. To find the eigenvector centrality, we will use the adjacency matrix as follows:
Let (A) be the adjacency matrix[5] of the network, where $(A_{ij} = 1)$ if there's an edge between (i) and (j), and (0) otherwise. The eigenvector centrality $(C_E\,(i))$ for node (i) is the (i)-th element of the eigenvector (v) corresponding to the largest eigenvalue λ of (A), satisfying: $Av = \lambda v$.

5) PageRank (named after the term *web page* and Google's co-founder Larry Page) is a variant of eigenvector centrality initially developed for ranking web pages. It can also measure the probability of random jumps between nodes. It accounts for the influence of a page's direct links and the quality (or PageRank) of the pages that link to it. The Page-Rank theory holds that an imaginary web user randomly clicking on links will eventually stop clicking. At any step, the probability that the person will continue following links is a

[5]In graph theory and computer science, an adjacency matrix is a square matrix used to represent a finite graph. The elements of the matrix indicate whether pairs of vertices are adjacent or not in the graph.

damping factor d. The PageRank (PR(i)) for node (i) is given by:

$$PR\,(i) = \frac{1-d}{N} + d\sum_{j\in B_i} \frac{PR\,(j)}{L\,(j)} \tag{7}$$

where d is the damping factor, N is the number of nodes, B_i is the set of nodes linking to i, and $L(j)$ is the out-degree of node j.

C. RISK ANALYSIS

In this work, we define the graph $G = (N, E)$, where N is the set of nodes which represent road junctions (or urban blocks), and E is the set of directed edges which represent roads bet-ween urban blocks/junctions. Directed edges also depict the dependency relationship between them. Each node is also characterized by a heat risk index H_{ni}, calculated based on population and real-time temperature data. The disruption transferred through this dependency can be illustrated by the combination of *Likelihood* ($L_{i,j}$), which represents the probability of heat-related risk affecting node N_j if node N_i is impacted, and the Impact ($I_{i,j}$), which quantifies the severity of the heat-related risk at N_j due to its dependency on N_i.

IPCC (Intergovernmental Panel on Climate Change) has given a general definition as ''the propensity or predisposition to be adversely affected''. According to IPCC [39], the majority of researchers define vulnerability as a function of exposure, sensitivity and adaptive capacity, where exposure ''refers to the direct danger of urban heat that affects the considered area, and manifests as the prevalence of very high temperatures'', strength quantifies ''human's reaction to high temperatures, and adaptive capacity ''refers to the ability to cope with, recover and adjust to the impacts of heat events'' [22].

1) THE ''LIKELIHOOD'' METRIC FOR EXTREME HEAT EVENTS

The *likelihood metric* is used to quantify the probability of extreme heat exposure at each node (road junction). The nodes used in the graph serve as geographical points where solar radiation and population vulnerability data are to be analyzed. The roads connecting these junctions – represented as edges in the graph – capture other relevant factors, such as road length.

The methodology presented utilizes satellite data provided by the Google Solar API[6] for real-time heat index calculations to reveal areas within cities that are more exposed to extreme temperatures. The solar radiation data, particularly the highest recorded solar radiation value for each junction, forms the basis for estimating the likelihood of heat vulnerability at each node. Higher solar radiation at a junction (an accumulation of solar radiation data for the surrounding area) indicates more significant exposure to heat, thus increasing the likelihood of heat-related risks. The more intense the solar radiation is, the more likely that node (junction) will be exposed to heat extremes.

[6]https://developers.google.com/maps/documentation/solar/overview

We can assign a metric, $L_{(N),}$ to each node N, representing the probability of heat-related risks based on climatic conditions. Specifically, the likelihood $L(N)$ at node (junction) N is defined as:

$$L\,(N) = \frac{S\,(N)}{S_{\max}} \tag{8}$$

where:

1) $S(N)$ is the highest solar radiation recorded at node N during the year 2023.
2) S_{max} is the maximum solar radiation value across all nodes in the network.

Solar radiation values are normalized between values 0 and 1, where 1 indicates the highest recorded exposure to heat and sun radiation, to ensure comparability between the nodes (junctions). This normalization ensures that each node's likelihood score is comparable within the network, underscoring the relative heat exposure across the urban area.

Heat exposure involves considering not just vulnerable groups but also exposure duration and general health status. Mapping this into our model is easily solved for our digital twin by aggregating the monthly flux data from Google Solar API. The Google Maps Solar API provides monthly flux data, which ''represents the sunlight exposure of a rooftop for each month of the year'' and ''shows the monthly yield of sunlight on a given surface, expressed in kWh/kW/month'' [32]. Give monthly flux averages that surpass the heatwave thresholds, defined statistically and empirically for Vienna for temperatures above 30-32 degrees Celsius, it is easy to consider that these durations can be marked as ''high duration'' and ''detrimental for general health''.

2) THE ''IMPACT'' METRIC FOR AFFECTED JUNCTIONS

The characteristics of the vulnerable population groups represent the impact of heat exposure vulnerability at each node. We can identify vulnerable population groups as those that include the elderly (people over 65 years of age), infants (people under 2 years of age), and people with pre-existing medical conditions (i.e., heart or lung diseases) [19]. The impact metric is based on the density and characteristics of vulnerable populations near each road junction (node). We will use the available demographic data to assign each node a population vulnerability score, $I(i)$. This metric represents the concentration of high-risk individuals near each node/junction. The impact score $I(i)$ for each node i is computed based on the total population of vulnerable age groups around that node, specifically aggregating individuals under 9 and over 60. This score is then normalized by dividing by Pmax, the maximum vulnerable population across all nodes, yielding a value between 0 and 1. The formula for calculating the normalized impact score $I(i)$ for each node i is as follows:

$$I\,(i) = \frac{P\,(i)}{P_{\max}} \tag{9}$$

where:

1) P(i) represents the vulnerable population score for node i, which is determined based on the number of individuals in vulnerable groups near that junction.
2) Pmax is the maximum population vulnerability score across all nodes.

First responder forces' deployment and emergency planning efforts can be prioritized based on the impact of the metric for each node (road junction). Nodes with high impact scores represent locations where extreme heat could lead to serious health implications, making them prime areas for preventive actions and rapid emergency response. In that view, this metric allows for areas with vulnerable populations to receive due consideration and support equity in emergency health response.

3) THE "RISK" METRIC FOR HEATWAVE SCORING

A 'heatwave risk score' is calculated by normalizing, scaling, and weighting data for likelihood and impact metrics above. Normalization for both metrics involves dividing observed values by maximum observed values across all nodes.

The final risk score for each city block and junction i combines these normalized scores using a geometric mean, defined explicitly as:

$$Risk\ (i) = \sqrt{L\ (i) * I(i)}$$

The geometric mean is selected because it appropriately penalizes extremes in both scores, thus ensuring balanced consideration of both exposure and impact [33], [34]. We will discuss the uses and justification of the geometric mean in section IV) CALCULATING CITY BLOCK RISK below.

IV. EMERGENCY INCIDENT RESPONSE PLANNING DU-RING HEAT EVENTS – VIENNA CITY CENTER CASE

To validate the effectiveness of our approach regarding the optimal response strategies in heat-vulnerable areas, such as urban regions with high elderly populations and limited shade cover, we examined the case of the Vienna city center. Bet-ween 1961 and 1990, the city of Vienna experienced an average of 9,6 days of extreme heat per year, with that number rising to 15,2 per year in 2010 [20]. More importantly, during the heatwave of 2003, Vienna experienced an overall of 44 extreme heat days, resulting in 180 deaths [21]. This section will describe the data set used in our analysis and the proposed methodology.

A. DATA SOURCE

1) VIENNA CITY CENTER POPULATION DATA

The data set used for the population forecast for the years 2014 to 2024 as per the Vienna census districts, is published by the City of Vienna (Vienna Municipal Administration). It is licensed under the City of Vienna's Open Government Data, which are licensed under a "Creative Commons Attribution 4.0 License" (CC BY 4.0). The City of Vienna (a statutory city itself) is divided into 23 named city sections or districts. Our focus is on district number I, with district code 901 and named as "Innere Stadt".

Based on the information provided in the dataset, the Vienna City center [District number: I, district code: 901XX, sub-codes: 90101 to 90107] as of 01.01.2024 had a total population of 15.390, of which 1826 were between the age 0 and 9 and 5055 where above the age of 60. The IPCC [Intergovernmental Panel on Climate Change] identifies these age groups as vulnerable. As per "The Urban Heat Vulnerability Map of Vienna," [22] the (heat) vulnerability is defined as a function of three factors:

1) *Exposure* refers to the direct hazard posed by urban heat affecting the designated area. Urban heat is characterized by persistently high temperatures across the city, known as "heat islands" or hot spots. The intensity of these hot spots can vary depending on various factors specific to the area and its built environment" [22].
2) *Sensitivity* refers to "human response to high temperatures, which depends on individual factors such as age, existing health conditions, and more. Sensitivity is evaluated by identifying individuals – in this scenario, the age groups previously identified - within a specific area most impacted by extreme heat" [22].
3) *Adaptive Capacity* refers to the ability to "manage, recover from, and adjust to the effects of heat events" [22]. In simple terms, adaptive capacity is the ability of systems within the urban environment to adapt and respond to changes, such as extreme heat.

2) VIENNA CITY CENTER ROAD NETWORK

Fetching road data is a simple process. There are several ways to achieve this, such as using open data and open street maps. To prepare the data for graph-based analysis, we can use the OSMnx [23] library to transform the data into x,y (lat and long) coordinates. These coordinates are used by several other programs and platforms, including Google Maps and Google Earth, providing extended functionality, availability, and versatility.

3) VIENNA'S URBAN DEVELOPMENT DATASET

The "Realnutzungskartierung ab 2007 Wien" [31] dataset offers a comprehensive overview of Vienna's land use derived from aerial imagery, supplemented by additional factual data. The dataset provides three usage categories for a total of 32 distinct categories in housing, green spaces, and transport, including health and emergency organizations. The hierarchical organization of these categories into the three levels provides a detailed framework for analyzing urban land usage. The detailed analysis of this dataset can provide insights related to the location of the emergency health services, their proximity of the vulnerable population to healthcare facilities and first responders, as well as analysis of the urban layouts and green spaces.

4) GOOGLE SOLAR API

Google Solar API [32] is part of Google's Environmental Insights Explorer (EIE) is a set of tools and data to estimate

the solar potential of rooftops across various geographic areas, and civil authorities, organizations, and urban planners can use to identify areas with high solar radiation. The Google Solar API [32] provides solar data for hundreds of millions of buildings world-wide, with imagery quality varying from HIGH (e.g., 10cm) to LOW (e.g., 30cm or worse). ''Data layers'' is one of the available endpoints in the Solar API. Each image layer provides unique information that, when combined, creates a com-prehensive picture of the rooftop. By using the Flux imagery, the level of sunshine (solar radiation) that hits the roof can be provided on a monthly and yearly basis. It accepts requests for three endpoints—

- buildingInsights, which returns insights about a building's location, dimensions, and solar potential;
- dataLayers, which provides URLs for raw solar information datasets covering an area around a location; and
- geoTiff, which fetches rasters with encoded solar data including a digital surface model, aerial imagery, annual and monthly flux maps, and hourly shade.

The API defines sunniness as relative sunlight exposure across a roof and reports sunshineQuantiles in 11 buckets (deciles) of annual sunniness values, with intermediate segments receiving values accordingly. Solar information is encoded in GeoTIFF rasters (returned via dataLayers), each comprising a matrix of pixels holding discrete (thematic) or continuous data across one or multiple bands, with bit depth (e.g., 8-bit) determining the range of values.

Flux maps (annual sunlight in kWh/kW/year) account for location-specific hourly irradiance grids, weather patterns (cloud cover), shading from obstacles, and roof orientation. Coverage spans hundreds of millions of buildings worldwide, with imagery quality classified as HIGH (0.1 m/pixel low-altitude aerial), MEDIUM (0.25 m/pixel high-altitude aerial), and BASE (0.25 m/pixel satellite, experimental).

5) DATASET PREPROCESSING

The preprocessing module begins by traversing the Google Solar API data to ingest all block-level CSVs, computing the 'SolarRadiation' values per location (lat,lon); it then applies a linear min–max transform to rescale these maxima into a uniform [1, 10] range.

Next, the digital twin standardizes the raw demographic data for the Vienna population, and loads the Vienna Road network JSON, to use its latitude and longitude to call extract ''general,'' ''children under five,'' and ''elderly 60 plus'' density figures. An impact metric is derived as the arithmetic mean of the children and elderly densities, while the general density values undergo the same [1, 10] min–max normalization.

In the integration phase, each node's scaled radiation and scaled population density values are combined via a geometric mean to produce a 'likelihood' score, and both 'likelihood' and the precomputed 'impact' metric are injected back into the node records. Finally, the code iterates over every edge in the graph, computes the average of the likelihood and impact values of its source and destination nodes, annotates the edge with these averages, and writes out the enriched JSON network to proceed with path and risk calculation and analysis.

B. METHODOLOGY

This section details the methodology for assessing risk in Vienna's city center by integrating solar radiation data with population vulnerability metrics. The focus is on producing actionable risk assessments of routes towards high-risk areas (i.e. maximum heatwave impact and high percentage of vulnerable population). This, tailored to emergency response planning and public health initiatives, will ensure targeted interventions in selected urban blocks.

1) DATA ACQUISITION AND PREPARATION

1) **Road network**. Data regarding Vienna's city center's road network (roads and junctions) are transposed into a graph $G = (V,E)$, where V is the set of road junctions, and E is the set of roads.
2) **Population vulnerability data**: Age-based demographic information is used to identify vulnerable populations in Vienna's city center urban blocks and map them with the relevant junction coordinates. Information on several age groups is retrieved for each city block.
3) **Solar radiation data**: Solar radiation data retrieved for every month of 2023 for city center's urban blocks.

2) CALCULATING THE VULNERABLE POPULATION PER COORDINATE

We calculated the vulnerable population using the data from *''VIE-Bevölkerungsprognose 2014 bis 2024—Wiener Zählbezirke''* obtained from Opendata Vienna. Based on the IPCC's recommendation, the age groups 0-14 and 60+ were identified as the most vulnerable to extreme heat. We then fetched the vulnerable population data for the coordinates corresponding to each junction and mapped the vulnerable population with road junctions.

For each set of coordinates within the city center blocks, we calculate the total number of vulnerable populations, which is the summary of the individuals per vulnerable age group. The population counts for the vulnerable age groups are summarized to give the total number of vulnerable individuals at that location. The total vulnerable population *Pvuln* at a given coordinate is calculated as:

$$P_{\text{vuln}} = AGE_{00_02} + AGE_{03_05} + AGE_{06_09} + AGE_{60_74} + AGE_{75+} \quad (10)$$

where: AGE_{xx_xx} corresponds to the various vulnerable age groups.

3) EXTRACTION OF MAX SOLAR RADIATION VALUES

For each city block, a folder containing the relevant date is created, which incorporates all monthly records of solar

radiation values. The goal is to identify the maximum solar radiation value for the year 2023, which represents the peak solar radiation. In that way, we have an estimation of the heat vulnerability index. The maximum solar radiation SR_{max} is calculated as the maximum of all values:

$$SR_{\max} = \max(SR_1, SR_2, \ldots, SR_{12}) \tag{11}$$

where: SR_{1to12}, are the solar radiation values for each month of the year.

To facilitate comparisons between different city blocks, the solar radiation values are then normalized between the values 0 and 1, so the block with the lowest solar radiation is assigned a value of 0, and the block with the highest solar radiation is assigned a value of 1. All other values are scaled accordingly between the two extremes. For each block k, the solar radiation value v is normalized as per the following formula:

$$SR_{\text{norm}} = \frac{v - SR_{\min}}{SR_{\max} - SR_{\min}} \tag{12}$$

where:

- v is the highest solar radiation value for a given block.
- $SR_{\min}$ is the minimum solar radiation value across all blocks.
- $SR_{\max}$ is the maximum solar radiation value across all blocks.

Finally, the result is rounded to two decimal places.

4) CALCULATING CITY BLOCK RISK

The normalized solar radiation values and the population vulnerability data (i.e., the appropriate age groups) are combined to retrieve the risk posed by these two factors for each city block and junction. The risk R, for each city block is calculated using the geometric mean of these two values calculated in the previous steps:

$$R = \sqrt{SR_{\text{norm}} \times P_{\text{vuln}}} \tag{13}$$

We chose to use the ''geometric mean'' since we needed to combine different variables (in this case, radiation and population) to give us a balanced view. This approach [33], [34] is particularly useful when values are normalized between 0 and 1, as it scales down extreme values and prevents them from dominating the average. Furthermore, the usage of the geometric mean, allows for the combination of values that are by default distinct in nature and measured on different scales, with neither factor dominating the results. Additionally, this approach allows for the combination of different risk factors that can summarize each other's effects, rather than a simple addition.

We adopt the geometric mean (multiplicative) aggregation in Eq. 13 to combine the hazard, exposure, and vulnerability factors. This choice ensures that the overall risk index is **non-compensatory**, meaning a deficiency in any one component strongly diminishes the total risk score. In contrast to a simple sum, a geometric mean does not allow one extremely high factor value to fully outweigh a very low value in another factor [36]. This property aligns with standard risk paradigms where risk is conceptually the product of hazard, exposure, and vulnerability [37].

Using a geometric mean means to average the log-transformed inherent values [33], [34]. This formulation preserves the interpretation of the index on the original scale in the context of a city scale analysis and inherently penalizes imbalanced contributions across factors with multiple criteria – for example, improving all factors modestly yields a higher risk reduction than dramatically improving just one factor [37]. Eq. 13 variables are assigned equal weight (i.e., equal exponents in the multiplicative formula), reflecting an assumption that hazard, exposure, and vulnerability are equally important in the absence of evidence to the contrary. This equal-weight approach is a common, neutral baseline in composite indicator design [37].

We acknowledge that different weighting schemes could be employed if justified; for instance, planners might weight vulnerability higher if social factors are deemed paramount in a given city. Our framework is flexible in this regard. In this study, however, sensitivity tests showed that moderate weight adjustments did not substantially change the ranked risk outcomes, so we retain equal weights for clarity and to minimize bias.

This approach ensures a balanced risk metric where the overall risk is not dominated or influenced at disproportional levels by either factor. The results are then incorporated into the city's road network data, and each node is associated with the relevant cartesian coordinates (latitude and longitude). We will use this data to calculate the cumulative dependency risk for each node. The overall methodology is described in the following algorithm.

Validation of the metric is done by comparing Risk results with predicted emergency response locations and intervention points from recorded historical responses by experts, specifically during Vienna's extreme heat events. First results demonstrate consistency with historical knowledge.

5) REAL-TIME NODAL RISK UPDATES

Up until this point, the methodology integrates static demographic (vienna opengov.data) and solar radiation data (Google Solar API) within the graph model of the city to identify critical urban junctions and city blocks and optimize response routes for extreme heat events.

However, to effectively capture the inherent complexity and dynamism of urban road networks, a digital twin must also include explicit mechanisms to dynamically update node risk profiles and adjust routing recommendations are necessary. The PANTHEON model iterations can incorporate real-time traffic and congestion data from google maps per road request, obtainable through the google maps API. This enables the ability for continuous recalculations of nodal risks and optimal paths as traffic conditions evolve, should a need for a real-time update is required.

Specifically, PANTHEON adapts the graph analytics algorithms by automatically adjusting node vulnerability scores

Algorithm Process_Vulnerability_Data

1. INITIALIZE *base_directory, input_json_path, output_json_path*.
2. LOAD the road network data **from** *input_json_path*.
3. FIND the highest SolarRadiation value for each folder (representing city blocks over different months):
a. **FOR EACH** folder in *base_directory*:
i. **SET** (highest_value $= -\infty$).
ii. **FOR EACH** CSV file in the folder:
Read the SolarRadiation column.
FIND the maximum value in the file:
[max_value = max (SolarRadiation)]
UPDATE (highest_value) if:
[max_value > highest_value]
iii. **STORE** (highest_value) for the folder.
4. NORMALIZE SolarRadiation values:
a. **FIND** (max_radiation) and (min_radiation) across all folders:
[max_radiation = max (highest_values) , min_radiation = min (highest_values)]
b. **FOR EACH** folder:
i. **CALCULATE** the normalized SolarRadiation value $E(i)$:

$$[\mathrm{E\,(i)} = \frac{\text{highest_value (i)} - \text{min_radiation}}{\text{max_radiation} - \text{min_radiation}}]$$

ii. **STORE** $E(i)$ for each folder.
5. PROCESS the road network data:
a. **FOR EACH** node in the road network:
i. **FETCH** vulnerable population data for the node coordinates using *find_vuln_ppl(coordinates)*.
ii. **CALCULATE** the total population risk $P(i)$ by summing the values of all vulnerable age groups:
$[\mathrm{P\,(i)} = \mathrm{AGE_00_02} + \mathrm{AGE_03_05} + \ldots + \mathrm{AGE_75+}]$
iii. **CALCULATE** the normalized impact value $I(i)$:

$$[\mathrm{I\,(i)} = \frac{\mathrm{P\,(i)}}{\mathrm{P_{max}}}]$$

iii. **CHECK** if the node's folder name exists in the normalized SolarRadiation data:
IF EXISTS, calculate the likelihood $L(i)$ using the geometric mean:

$$[\mathrm{L\,(i)} = \sqrt{\mathrm{E\,(i)} \times P\,(\mathrm{i})}]$$

UPDATE the node's *likelihood* value in the JSON structure with $L(i)$.
iv. **IF NOT EXISTS**, **PRINT** an error message and exit.
6. SAVE the updated road network data:
a. **WRITE** the updated JSON structure to *output_json_path*.
7. END ALGORITHM.

based on google maps API value changes in traffic patterns to enable accuracy and reliability in real-world emergency response scenarios. Although this ability exists, the following results from scenario executions do not use real-time traffic since tangible scenarios would require an existing heatwave to incorporate; something that is considered both infeasible at the time period of writing this paper as well as out of scope for the goals of the involved first responders.

V. RESULTS

This section presents the results of implementing our approach into a real-world case scenario. In this approach, we applied a dependency-based risk analysis approach aiming to support and enhance emergency response and first-responding planning in case of an extreme heat event occurring in Vienna's city center. For this, we have incorporated dependency risk assessment methodologies – focusing on interconnected systems – in interconnected systems, integrating solar radiation data (as provided by the Google Solar PAI), population vulnerability metrics (i.e., vulnerable age groups), and network centrality measures. As a result, we managed to identify critical road junctions (nodes) and optimal paths (edges) which can assist city planners and first responders to prioritize emergency response efforts. These insights can result in targeted interventions that align with areas of highest exposure and cumulative risk, particularly around crucial infrastructure nodes such as health services.

A. HEAT AND VULNERABLE POPULATION DISTRIBUTION PER ROAD JUNCTION

At this step, the digital twin models heat distribution across the city by combining solar radiation data and population vulnerability metrics and normalized as described in the methodology. The results highlight areas with the highest risk of heat exposure and vulnerable populations. By combining the retrieved data on solar radiation (as derived from the Google Solar API) and information on vulnerable populations based on the recommendations and findings of the [22], we calculate the risk score for each road junction based on its tangent city blocks. Our method involves determining a heat exposure score based on solar radiation intensity balanced across blocks to allow for comparison. For each node we identified the vulnerable groups - groups under 2 years old and over 60 -, and superimposed to each road junction.

Figure 2 presents a color-coded visualizations that depicts the intensity of risks (vulnerability levels and heat exposure) across urban zones in Vienna center. This output is given to first responders to facilitate quicker interpretation and decision-making by emergency responders and planners.

As demonstrated in other studies, such as [24] on walkability scores, integrating multiple environmental and demographic factors into a spatial risk framework helps pinpoint areas needing enhanced planning for public health and emergency response.

For each node, a vulnerability score is calculated and then visualized, as depicted in Fig.1, which indicates which surrounding areas are more susceptible to effects resulting from extreme heat events. The current resolution was sufficient to identify major risk clusters, as those tend to cover multiple blocks and thus emerge even in coarse data. We have observed that densely populated areas, where elevated numbers of vulnerable populations are correlated with high solar exposure, created high-risk clusters.

B. JUNCTION CRITICALITY BASED ON BLOCK RISK

We assessed the criticality of each node by overlaying the impact and likelihood factors to prioritize nodes in emergency planning.

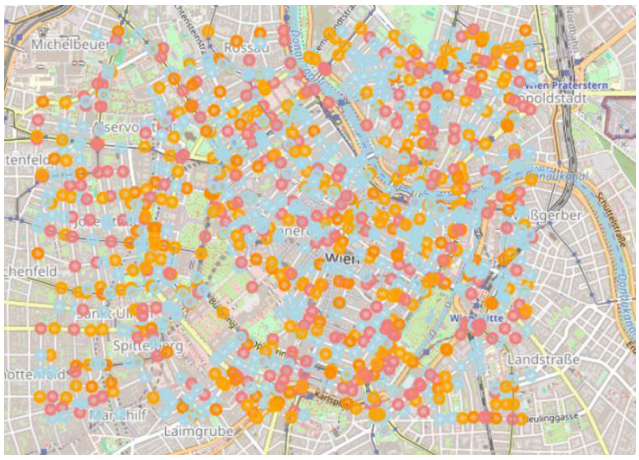


FIGURE 1. An overview of road junctions with nodes depicting their dependency risk. Risk value < 29 = 'lightblue', 30 <= p_value < 49 = 'lightcoral', 50 <= p_value < 79 = 'orange', 80 <= p_value < 100 = 'red'.

This step identifies critical junctions within the road network by evaluating block-level risks derived from normalized heat and population data. These results prioritize areas requiring intervention based on cumulative risk measures. We have exploited several centrality metrics to identify critical junctions within the road network. More specifically, to underline the influence and reach of each junction (node), we used the PageRank and Eigenvector, which are based on the node's connections to highly connected neighboring nodes. The importance of the node acting as a bridge between other nodes was particularly valuable in determining network vulnerability points. To measure this, we have identified high betweenness scores. This approach uses the method by [4] for assessing congestion in transportation networks, where dependency paths among road junctions were calculated to prioritize infrastructure improvements and improve resilience.

By analyzing relevant results, we identify nodes that are both highly connected and strategically located within the network. Identifying these critical nodes enables the effective and efficient allocation of emergency response resources, thus enhancing their effectiveness across the network. Our research shows that, to allow a prioritized emergency response in adverse conditions and areas with heightened risk and critical infrastructure presence, a centrality-based approach best fits for analyzing node connectivity and significance within a complex network. Table 2 presents an example of calculated importance scores for city junctions located at the given coordinates. These metrics quantify the importance of each node based on its position and connectivity within Vienna's city road network, as calculated in the road network graph. The centrality scores reflect the relative significance of each node (which represents a city block) in the overall network.

The geodetic coordinates (lat, long) have been calculated for each road junction (node) and are depicted here in parentheses.

TABLE 2. Extract from the centrality results.

Node	Eigenvector	Betweenness	Degree
1-(48.2068359,16.3558556)	0.006	22,131.138	2
2-(48.2076973,16.3553748)	0.009	17,590.179	2
3-(48.214422,16.3582651)	0.013	48,881.755	2
4-(48.2135166,16.3571308)	0.004	3,550.194	2
5-(48.2133903,16.358084)	0.005	71,848.94	3
6-(48.2124349,16.3588007)	0.011	40,857.588	2
7-(48.2125633,16.3578329)	0.01	73,570.855	2
8-(48.212794,16.3560356)	0.004	66,720.351	1
9-(48.2118784,16.3558256)	0.023	143,861.021	2
10-(48.2117779,16.3566008)	0.018	71,643.351	2
[…]	[…]	[…]	[…]

The geodetic coordinates (lat, long) have been calculated for each road junction (node) and are depicted here in parentheses.

From the evaluation of the relevant results, we can identify road junctions that are not only subjectable to heat risks but also are important for the urban network. For example:

1) High betweenness centrality means that the node serves as a bridge between key areas (other nodes) and is vital for the effective flow of emergency resources and first responders.
2) Eigenvector centrality means that the road junction acts as a highly influential node within the network, due to its connection to other important nodes. This measure is critical when determining the cascading effects on surrounding nodes and areas.

C. CUMULATIVE DEPENDENCY RISK FOR EACH NODE

This step calculates the cumulative dependency risk for each node by aggregating risks along interconnected paths, as explained in the dependency risk model. The results provide insights into potential cascading effects across the urban network. The cumulative dependency risk is calculated for each node from the combination of the likelihood of heat exposure, as calculated from Google Solar API, with the impact this exposure may have, as expressed in the vulnerable population density. Our methodology addresses issues arising from risk assessment calculations across multiple paths, identifying routes with higher dependency risks because of the node's connections. Key observations are:

1) *High-risk Dependency Paths:* Routes and their dependent road junctions, in which a single disruption can impact multiple critical areas.
2) *Scalable Risk Analysis:* By scaling our model across nodes, we were able to create a risk mapping that identifies risks related to specific junctions and the possibility of cascading effects across the network.

Furthermore, we evaluated our methodology by examining a specific node, namely node 687, as our focal point. This node represents a point where emergency health services are provided and where first responders may initiate their patrols. The choice of Node 687 was because it can effectively demonstrate our methodology's capability to reveal cascading risks. In this example, the dependency risk between

nodes is calculated through a weighted sum of individual risks along connected nodes. In that way, it is possible to identify potential cascading effects in areas with high heat exposure and large numbers of vulnerable populations. Cumulative risk associated with every path is measured by aggregating the likelihood of heat exposure with the vulnerability metrics at each node. The paths originating from Node 683 illustrate how high dependency risks can propagate through multiple junctions, affecting broader areas if not mitigated. Using the dependency risk equation (2) we calculated the cumulative risks for paths originating from Node 683 as follows:

$$Dependency\ Risk(DR) = \sum_{i=1}^{n} (L_i \cdot I_i) \qquad (14)$$

where:

1) L_i: is the likelihood of heat-related risk for node i.
2) I_i: is the impact based on the vulnerable population at node i.

Node 683 is identified as a critical "cold spot", a strategic location for ambulances and first responders to provide shade and support during heatwaves. Node 683 is identified as a critical junction, prioritized for emergency planning during heatwaves. Table 3 provides an example of risk paths calculated by aggregating the risk scores of junctions along possible routes. Each node risk is determined by integrating solar radiation and population data, representing a city block's combined vulnerability.

This approach provides emergency response planners and first responders with the flexibility to adapt their intervention plans according to the operational requirements. Rapid response can be ensured through the selection of the shortest paths, while longer paths allow for the comprehensive coverage of a large number of high-risk areas. In that way, emergency resources (limited by default) are effectively and efficiently allocated to address the increased impact via the most valuable locations within the urban network.

TABLE 3. Cumulative dependency risks when Node 683 is the source node.

Path	Cumulative Dependency Risk	Path Size
683-(48.2053571,16.3886194) -> 648-(48.205181,16.3884332)	**3.15**	2
683-(48.2053571,16.3886194) -> 673-(48.2057073,16.3879553)	**2.82**	2
683-(48.2053571,16.3886194) -> 673-(48.2057073,16.3879553) -> 131-(48.2050556,16.3874845)	**3.18**	3
683-(48.2053571,16.3886194) -> 673-(48.2057073,16.3879553) -> 131-(48.2050556,16.3874845) -> 678-(48.2042226,16.3886713)	**3.2088**	4
683-(48.2053571,16.3886194) -> 673-(48.2057073,16.3879553) -> 131-(48.2050556,16.3874845) -> 685-(48.205542,16.3863549)	**3.3528**	4
683-(48.2053571,16.3886194) -> 673-(48.2057073,16.3879553) -> 131-(48.2050556,16.3874845) -> 685-(48.205542,16.3863549) -> 1021-(48.2057668,16.3856033)	**3.612**	5
683-(48.2053571,16.3886194) -> 673-(48.2057073,16.3879553) -> 131-(48.2050556,16.3874845) -> 685-(48.205542,16.3863549) -> 1021-(48.2057668,16.3856033) -> 1018-(48.205811,16.3854538)	**3.77328**	6
683-(48.2053571,16.3886194) -> 673-(48.2057073,16.3879553) -> 131-(48.2050556,16.3874845) -> 685-(48.205542,16.3863549) -> 1021-(48.2057668,16.3856033) -> 1018-(48.205811,16.3854538) -> 922-(48.2047605,16.3846742)	**3.782496**	7
683-(48.2053571,16.3886194) -> 648-(48.205181,16.3884332)	**3.15**	2
683-(48.2053571,16.3886194) -> 673-(48.2057073,16.3879553)	**2.82**	2
683-(48.2053571,16.3886194) -> 673-(48.2057073,16.3879553) -> 131-(48.2050556,16.3874845)	**3.18**	3

D. OPTIMAL PATHS FOR EMERGENCY RESPONSE

At this step, the digital twin identifies optimal paths for emergency response by integrating node-level risks and dependency chains. The results propose prioritized paths based on user-defined criteria, such as path length or risk coverage. Using the calculated dependency risk data, we can then identify optimal paths for first responders and emergency planning, that address exposure to high-risk zones, and at the same time maximize the coverage of vulnerable populations. We have identified the following cases:

1) Paths that connect multiple high-risk nodes [please refer to Use case#1 below]
2) Routes that provide the shortest high-risk paths allowing for rapid access to specific critical points. [please refer to Use case#2 below]

By identifying these two use cases, we introduce the ability for first responders to utilize extensive coverage of vulnerable populations while minimizing costs associated with the chosen path. This approach improves coverage and reduces emergency response time, enabling the effective protection of vulnerable populations. Focusing on high cumulative dependency risk nodes helps strategically position relevant resources and incorporate proactive measures to efficiently and comprehensively address urban heat vulnerabilities.

Quantitative performance metrics, including measurable reductions in emergency response times and improvements in the coverage of vulnerable populations, cannot be added explicitly without extended historical reference points where the simulation is run in parallel to real-world emergencies. To this end and to validate results, first responder experts compare between predicted response patterns generated by our model and historically recorded emergency responses and check for discrepancies. Results demonstrate tangible benefits, such as reduced response times in planning and more efficient resource allocation.

The use of first responder expertise contributes to explicitly distinguishing between methodological advancements—such

Path:
683-(48.2053571,16.3886194) -> 673-(48.2057073,16.3879553) -> 131-(48.2050556,16.3874845) -> 685-(48.205542,16.3863549) -> 1021-(48.2057668,16.3856033) -> 1018-(48.205811,16.3854538) -> 922-(48.2047605,16.3846742) -> 1085-(48.2044656,16.3843064) -> 1084-(48.2044558,16.3839403) -> 1002-(48.2044246,16.3836854) -> 1003-(48.2044725,16.3832341) -> 882-(48.2044168,16.3830526) -> 830-(48.2031131,16.3815651) -> 49-(48.2017984,16.3799428) -> 708-(48.2017517,16.3801049) -> 815-(48.2016134,16.3802076) -> 816-(48.2010226,16.3793335) -> 814-(48.2005419,16.3786333)
Risk = 3.8
Junction length = 18

FIGURE 2. Use case scenario #1. The longest risk path, originating from Node 683 connecting nodes with highest risk.

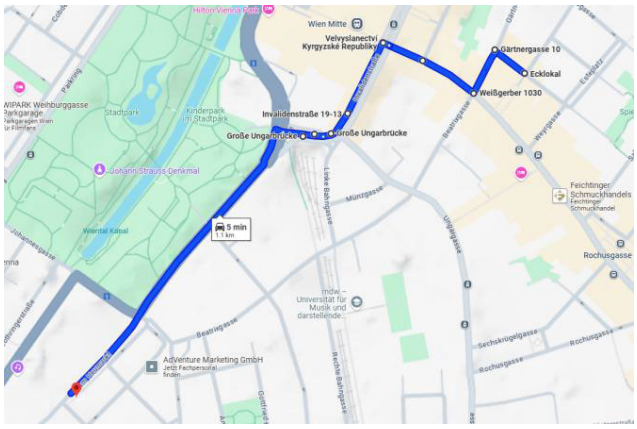


FIGURE 3. The graphical representation of the highest and longest risk path calculated for Node.683 acting as source node.

as integrating spatial dependency analytics with real-time environmental data—and applied insights from expert knowledge. Given that first responders validate the methodology through data and output review, this underscores the technical innovation of our model and its practical impact for enhancing urban resilience.

Use case #1 – Highest and Longest Risk Path

The results of the relevant calculations are depicted in Fig.2 below, while the graphical representation is depicted in Fig. 3. Results represent risk paths originating from Node 683, calculated by aggregating risk scores of individual nodes along each route. **Results identify junctions that serve as critical hubs in the network**, either due to their connectivity, influence on surrounding areas, or importance in mitigating cascading risks. City planners and emergency responders can use these insights for prioritizing infrastructure improvements and resource allocation. By doing that, they can achieve max network resilience.

Use case #2 – Shortest and Highest Risk Path

The results of the relevant calculations are depicted in Fig.4 below, while the graphical representation is shown in Fig.5. As shown in these results, the cumulative risk coverage and path length constraints can be balanced in such a way as to provide optimal paths for emergency responders. These paths that connect multiple high-risk nodes ensure efficient coverage of critical areas in minimum response times. This analysis supports strategic planning for emergency scenarios during extreme weather events (such as heatwaves).

Path:
683-(48.2053571,16.3886194) -> 673-(48.2057073,16.3879553) -> 131-(48.2050556,16.3874845)
Risk = 3.1
Junction length = 3

FIGURE 4. Use case scenario #2. The shortest risk path, originating from Node 683 connecting nodes with highest risk.

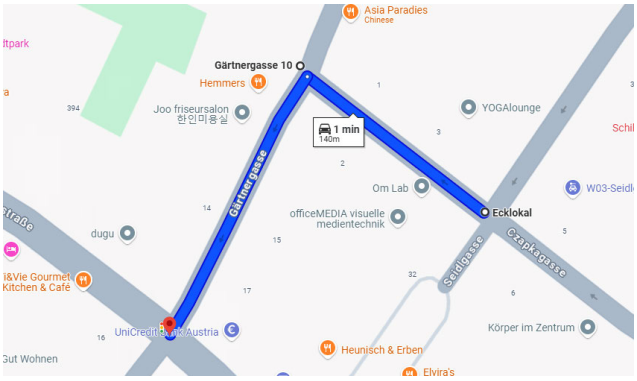


FIGURE 5. The graphical representation of the highest and shortest risk path calculated for Node.683 acting as source node.

TABLE 4. Global node-level correlation analysis (Spearman's P).

GLOBAL NODE-LEVEL CORRELATION ANALYSIS (SPEARMAN'S P)

Sensitivity pairs (u,d)	Spearman's P
Likelihood vs population density (scaled)	ρ = 0.662
Likelihood vs SolarRadiation (scaled)	ρ = 0.762
Impact vs population density (scaled)	ρ = 1.000
Impact vs SolarRadiation (scaled)	ρ = 0.070

GLOBAL EDGE-LEVEL CORRELATION ANALYSIS (SPEARMAN'S P)

Sensitivity pairs (u,d)	Spearman's P
Likelihood vs population density (scaled)	ρ = 0.760
Likelihood vs SolarRadiation (scaled)	ρ = 0.640
Impact vs population density (scaled)	ρ = 1.000
Impact vs SolarRadiation (scaled)	ρ = 0.051

E. SENSITIVITY ANALYSIS

To examine the robustness of the risk assessment, we performed a sensitivity analysis on the heat hazard input. In our base scenario, the peak daytime heat index was [1-35 °C]; here we varied within the entire range, studying each value with a margin of ±5% (approximately ±2 °C) and computed the relevant dependency risk indices for Global node-level correlation analysis of sensitivity for each variable tuple used in these experiments *(u,d,) where u indicates an output metric, and d indicates the respective input analyzed against.*

Results depicted in Table 4 demonstrate that the resulting risk maps remain highly correlated with the environmental and spatial inputs (Overall Spearman's $\rho \approx 0.8$), confirming that the digital twin's risk estimations faithfully reflect underlying heatwave hazard and demographic vulnerability drivers. This strong monotonic relationship indicates that the model reliably identifies high-risk city blocks even under minor variations in heatwave intensity, ensuring robust sensitivity to environmental perturbations.

Likelihood (the geometric mean of radiation and density) shows strong alignment with both inputs, while Impact (average of vulnerable-group densities) is perfectly rank-correlated with overall population density ($\rho = 1.000$), which is obvious since every node's impact ranking is identical to its general density ranking. Again, Impact shows no meaningful association with SolarRadiation ($\rho \approx 0.07$), as expected since radiation does not enter the impact calculation. Averaging across edges yields very similar patterns: likelihood remains strongly tied to both inputs (slightly more to density this time), and impact is again perfectly aligned with population density, but uncorrelated with radiation.

Likewise, Vulnerable-group densities are scaling proportionally to general densities across all nodes. In other words, areas with higher overall population density also have proportionally higher densities of the considered vulnerable groups, with no reordering.

Since likelihood $= \sqrt{(\text{scaled_radiation} \times \text{scaled_density})}$, likelihood's ordering naturally lies between the two inputs. The fact that $\rho \approx 0.76$ vs. radiation and 0.66 vs. density confirms that both factors contribute meaningfully—and neither completely dominates.

This sensitivity analysis confirms that our digital twin framework can effectively calculate and discriminate risk levels for urban blocks in the context of heatwave scenarios and meaningfully correlate data to map risk both between locations and in clusters.

VI. CONCLUSION

This study presents a novel methodology for enhancing heatwave response planning in urban centers using a digital twin model that integrates solar radiation data, population vulnerability metrics, and graph-based network analysis. We aim to identify high-risk urban blocks by focusing on node-level risks. In that view, we can use efficient resource allocation and rapid access to high-risk areas by providing optimized response paths to first responders.

This study provides tangible benefits from the practical application of the findings into risk prioritization and route optimizations for the optimized urban heatwave response preparedness and planning. First responders (e.g., an ambulance) can utilize the proposed methodology to identify paths that maximize risk coverage while adhering to operational or other limitations, such as ensuring that the patrol route does not exceed a duration of 15 minutes, by – for example – focusing on paths with a depth of 4 to 7 nodes. Furthermore, this methodology allows for strategically placing appropriate resources at nodes within the critical paths allowing for rapidly reaching any point within their patrol area. The dual focus of this methodology ensures that emergency resources are both targeted toward the most vulnerable areas and readily available for rapid deployment in case of emergencies.

The results demonstrate the effectiveness of this framework in prioritizing emergency planning. Key findings include the identification of critical junctions with high cumulative dependency risks and the generation of optimal paths tailored to user-defined constraints, such as path length and coverage of vulnerable populations. For example, we identified node 683 as a critical ''cold spot.'' Our model can guide first responders and city planners to strategically place the appropriate resources and, at the same time, suggest appropriate intervention strategies during extreme weather events, such as heat waves. City planners and public health authorities can benefit from this framework with the introduction of targeted interventions and minimizing health impacts on vulnerable populations.

The proposed approach is transferable to other urban settings, provided it is calibrated to local conditions. We anticipate that with appropriate Solar API data, city's specific data availability, infrastructure, and road characteristics, the framework could be adapted to improve heatwave planning in other cities.

Incorporating real-time data (such as environmental and demographic) and additional factors (such as transportation network disruptions) further expands the predictive capabilities of our model. The extension and expansion of our model can be achieved through its application to other cities or by simulating other extreme events (such as flooding). As a result, this methodology provides a robust foundation for urban resilience against heat waves and other climate-induced emergencies.

VII. IMPLICATIONS AND FUTURE WORK

Future work should explicitly address the implications of our findings within the context of European Union regulations and directives, particularly the EU Strategy on Adaptation to Climate Change (COM(2021) 82) and the revised Network and Information Security Directive (NIS2 Directive).

According to NIS2., which mandates comprehensive risk-based approaches to protecting essential entities, member states must ensure resilience of essential services such as healthcare and emergency infrastructure; our findings directly support these obligations by pinpointing critical urban nodes where targeted interventions during heatwaves would substantially improve emergency response effectiveness.

In terms of databases used, the Google API solar irradiance data, scale resolution and grid layers are already too detailed for the purposes of a block-level heat mapping. Uncertainty in data is of low risk due to the inherent high-analysis data provided by Google for the purposes of this project. Per-structure analysis and variance between daily values-per-build, are way more detailed than what is needed per day, city-level block mapping of a heatwave. This work could incorporate downscaled or locally measured solar radiation data for urban areas to minimize processing time. We suggest that integrating lower-resolution data while performing a location-specific calibration (e.g. using ground sensors), would be a useful next step to reduce overhead in calculations while keeping uncertainty too low levels.

Future research directions should also specifically aim at integrating real-time data streams from EU-supported environmental monitoring platforms (such as the Copernicus Earth Observation Programme) to enhance predictive accuracy for real-time emergency management decisions. Furthermore, a critical direction would be to expand the methodology's applicability to also include modeling compound events, as emphasized by the EU Adaptation Strategy (COM(2021) 82), to address scenarios where heatwaves intersect with other infrastructural disruptions such as power outages, transport disruptions, or water shortages, thus ensuring robust compliance with EU resilience and climate adaptation policies.

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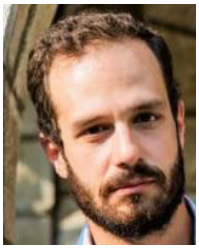
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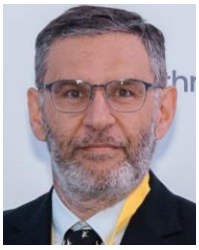
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